1 Introduction

The question of whether a closed manifold admits a metric of positive scalar curvature has been a central topic in differential geometry over the past decades and remains an area of active research. One part of obtaining an answer to this question consists in finding suitable obstructions to the existence of positive scalar curvature metrics. For example in dimension 2 the Euler characteristic of a closed manifold is such an obstruction by the Gauß-Bonnet Theorem and by the classification of surfaces this is in fact the only obstruction. However in higher dimensions this question becomes much more subtle. One way to approach this problem is by means of index theory of the spin Dirac operator. In 1980 Gromov and Lawson [**?**] have used this technique to identify a large class of manifolds, which do not admit positive scalar curvature metrics, namely so called enlargeable manifolds. We will give a precise definition of this notion below, from which we will easily derive the following two facts:

1. S^1 is enlargeable,
2. Products of enlargeable manifolds are enlargeable.

From this it follows at once that the n-Torus T^n does not admit a metric of positive scalar curvature for all dimensions $n \geq 0$, which had long been an open question. This example already highlights the importance of the concept of enlargeabilty, which, since its emergence, has been a subject of extensive research. More generally, the two facts above show that if M is enlargeable, $M \times S^1$ does not admit a metric of positive scalar curvature. There are two natural ways one may try to generalize this statement. Firstly, we can ask what happens if we replace M by any manifold, which does not admit a metric of positive scalar curvature. Naively one might expect the statement to still be true, nonetheless it turns out that there are counterexamples, see [??]. Secondly, we can ask what happens if the S^1-factor appears as the fiber of some non-trivial circle bundle over M. However, once again, this seemingly plausible expectation does not hold. Counterexamples to this second question have recently been constructed in [??] by relying on techniques from symplectic geometry. In the following we will provide an alternative construction for such counterexamples, employing only basic topological notions. Let us begin by recalling the notion of enlargeabilty.

Definition 1.1. A closed oriented Riemannian n-manifold M is called enlargeable if for all $\varepsilon > 0$ there exists a finite Riemannian covering $\hat{M} \to M$ together with an ε-contracting (that is ε-Lipschitz) map $\hat{M} \to S^n$ of non-zero degree.

Note that this notion does not depend on the particular choice of Riemannian metric on M, since for closed manifolds all metrics are in bi-Lipschitz correspondence. Our definition of enlargeabilty is adopted from [**?**] in contrast to the orginial definition in [**?**], where the coverings $\hat{M}$ were additionally required to be spin. The observation that enlargeabilty is an obstruction to the existence of positive scalar curvature metrics was first shown by Gromov and Lawson [**?**] in the case where M is spin. This was later generalized by Cecchini and Schick [**?**] to the non-spin case.

As already mentioned above, the simplest example for an enlargeable manifold is S^1. To see this consider the Riemannian covering $(S^1, ng) \to (S^1, g), z \mapsto z^n$, where g is the standard metric on S^1. Then the map $(S^1, ng) \to (S^1, g), z \mapsto z$ is $1/n$-Lipschitz and of degree 1. Since we can choose n arbitrarily large it follows that S^1 is enlargeable.

The following Lemma now provides two easy ways to construct new enlargeable manifolds out of given ones.

Lemma 1.2. *(i) The product of enlargeable manifolds is enlargeable.*

(ii) A closed oriented manifold M admitting a non-zero degree map onto an enlargeable manifold is enlargeable.

Proof. □

From Part (ii) of the above Lemma it follows immediately, that enlargeabilty is a homotopy invariant, since a homotopy equivalence of connected closed oriented manifolds has degree ± 1. In fact more is true, ...

2 The Construction

We will now construct counterexamples to the question raised in the beginning, that is circle bundles $E \to M$ over enlargeable M, where the total space E admits a metric of positive scalar curvature. In order to show the existence of such a metric we will employ the following Theorem.

Theorem 2.1 (Gromov, Hanke [**?**])**.** *Let V be a closed connected oriented smooth n-manifold with non-spin universal cover, $n \geq 5$. Let $\phi\colon V \to B\pi_1(V)$ be the classifying map and assume that there exists a closed oriented n-manifold V', which admits a positive scalar curvature metric, as well as a map $f\colon V' \to B\pi_1(V)$, such that*

$$\phi_*([V]) = f_*([V']) \in H_n(\pi_1(V);\mathbb{Z}).$$

(This includes the case $\phi_*([V]) = 0$.) Then V admits a positive scalar curvature metric.

Let N^{2n} be any $2n$-dimensional manifold with non-spin universal cover, $n \geq 2$. We could for example take $N := \mathbb{C}P^2 \times T^{2n-4}$ (hier vllt noch kurz Begründung reinschreiben). Consider the manifold

$$M := \mathbb{C}P^n \# T^{2n} \# N.$$

Since the collapse map $M \to T^{2n}$ has degree 1 and T^{2n} is enlargeable, it follows from Lemma 1.2 that M is also enlargeable. We will now construct a circle bundle $E \to M$ as follows. Let

$$c_1 \in H^2(M;\mathbb{Z}) \cong H^2(\mathbb{C}P^n;\mathbb{Z}) \oplus H^2(T^{2n};\mathbb{Z}) \oplus H^2(N;\mathbb{Z})$$

be the cohomology class corresponding to the first chern class $\alpha \in H^2(\mathbb{C}P^n;\mathbb{Z})$ of the canonical bundle $S^{2n+1} \to \mathbb{C}P^n$. We then define $E \to M$ as the unique S^1-principal bundle with first chern class equal to c_1. Now this is quite an abstract definition, so let us develop an intuition for how this bundle looks like. We claim that E is trivial over the summand $T^{2n}\#N$ and is given by the restriction of the canonical bundle $S^{2n+1} \to \mathbb{C}P^n$ over the $\mathbb{C}P^n$ summand. To make this precise let

$$i_1\colon (T^{2n}\#N) - D^{2n} \to M$$

be the inclusion. Then

$$\begin{tikzcd}
H^2(M;\mathbb{Z}) \arrow[r, "i_1^*"] & H^2((T^{2n}\#N) - D^{2n};\mathbb{Z}) \\
H^2(\mathbb{C}P^n;\mathbb{Z}) \oplus H^2(T^{2n}\#N;\mathbb{Z}) \arrow[u, "\cong"] \arrow[r, "\text{proj.}"] & H^2(T^{2n}\#N;\mathbb{Z}) \arrow[u, "\cong"]
\end{tikzcd}$$

commutes and therefore $i_1^*(c_1) = 0$. Again using the fact that S^1-principal bundles are uniquely determined by their first chern class we see that i_1^*E is indeed trivial. Similiarly if we denote by

$$i_2\colon \mathbb{C}P^n - D^{2n} \to M, \quad j\colon \mathbb{C}P^n - D^{2n} \to \mathbb{C}P^n$$

the inclusions, the diagram

$$\begin{tikzcd}
H^2(M;\mathbb{Z}) \arrow[r, "i_2^*"] & H^2(\mathbb{C}P^n - D^{2n};\mathbb{Z}) \\
H^2(\mathbb{C}P^n;\mathbb{Z}) \oplus H^2(T^{2n}\#N;\mathbb{Z}) \arrow[u, "\cong"] \arrow[r, "\text{proj.}"] & H^2(\mathbb{C}P^n;\mathbb{Z}) \arrow[u, "\cong", "j^*"']
\end{tikzcd}$$

commutes and thus $i_2^*(c_1) = j^*(\alpha)$. So the first chern classes of i_2^*E and j^*S^{2n+1} agree, hence they must be isomorphic bundles.

With this intuition in mind we now want to show the existence of a metric of positive scalar curvature on E by applying Theorem 2.1. In order to apply the Theorem we will show the following two claims.

Claim 1. E is totally non-spin, that is the universal cover of E is non-spin.

Claim 2. The projection $p\colon E \to M$ induces an isomorphism on π_1.

Granting these Claims for now let us see why they suffice for the assumptions of Theorem 2.1 to be satisfied. For this we would like to show that the classifying map $\phi\colon E \to B\pi_1(E)$ maps $[E]$ to $0 \in H_{2n+1}(\pi_1(E);\mathbb{Z})$, which can be seen by using Claim 2 as follows. Let $q\colon DE \to M$ be the disk bundle corresponding to E. Moreover let $\Phi\colon DE \to B\pi_1(DE)$ be the classifiying map for DE. Then the diagram

$$\begin{tikzcd} DE \arrow[r, "\Phi"] & B\pi_1(DE) \\ E \arrow[u, "\iota"] \arrow[r, "\phi"] & B\pi_1(E) \arrow[u, "B\pi_1(\iota)"'] \end{tikzcd}$$

commutes (hier noch Begründung rein), where the vertical maps are (induced by) the inclusion $\iota\colon E \to DE$. We claim that $B\pi_1(\iota)$ is an isomorphism. This follows from commutativity of

$$\begin{tikzcd} B\pi_1(E) \arrow[rr, "B\pi_1(\iota)"] \arrow[dr, "\cong", "B\pi_1(p)"'] & & B\pi_1(DE) \arrow[dl, "\cong"', "B\pi_1(q)"] \\ & B\pi_1(M) & \end{tikzcd}$$

where $B\pi_1(q)$ is an isomorphism by homotopy invariance and $B\pi_1(p)$ is an isomorphism by Claim 2. Thus ϕ factors over ι. Since DE is an oriented null-cobordism of E it follows that $\iota_*[E] = 0 \in H_{2n+1}(DE;\mathbb{Z})$ and so in particular $\phi_*[E] = 0 \in H_{2n+1}(\pi_1(E);\mathbb{Z})$ as we wanted to show. So together with Claim 1 all assumptions of Theorem 2.1 are satisfied, showing that E admits a metric of positive scalar curvature. We will now proof Claim 1 and Claim 2

Proof of Claim 1. Let $i\colon N - D^{2n} \to M$ be the inclusion. We have already established above, that E is trivial over the $T^{2n}\#N$ summand and so in particular

$$i^*E \cong (N - D^{2n}) \times S^1.$$

Note that if N is totally non-spin then so is $N - D^{2n}$ (hier auch evtl noch Begründung rein). Write $N_0 := N - D^{2n}$. Then the universal cover of i^*E is given by $\tilde{N}_0 \times \mathbb{R}$, where $\tilde{N}_0$ is the universal cover of N_0. Let $\pi_{\tilde{N}_0}\colon \tilde{N}_0 \times \mathbb{R} \to \tilde{N}_0$ and $\pi_{\mathbb{R}}\colon \tilde{N}_0 \times \mathbb{R} \to \mathbb{R}$ be the projections. We compute

$$w_2(T(\tilde{N}_0 \times \mathbb{R})) = w_2(\pi^*_{\tilde{N}_0} T\tilde{N}_0 \oplus \pi^*_{\mathbb{R}} T\mathbb{R}) = \pi^*_{\tilde{N}_0} w_2(T\tilde{N}_0).$$

Since $\pi_{\tilde{N}_0}$ is a homotopy equivalence and $\tilde{N}_0$ is non-spin it follows that $w_2(T(\tilde{N}_0 \times \mathbb{R})) \neq 0$, such that i^*E is totally non spin. Therefore M contains a totally non-spin submanifold of codimension 0 and is consequently itself totally non-spin by (hier vllt noch kurze Begründung reinschreiben) $\square$

Proof of Claim 2. Let $j\colon \mathbb{C}P^n - D^{2n}$ be the inclusion. Consider the following commutative diagram, where the rows are taken from the long exact sequences of homotopy groups of the respective fibrations and the vertical maps are induced by inclusions:

$$\begin{tikzcd} \pi_2(M) \arrow[r, "\partial_1"] & \pi_1(S^1) \arrow[r] & \pi_1(E) \\ \pi_2(\mathbb{C}P^n - D^{2n}) \arrow[u] \arrow[r, "\partial_2"] \arrow[d, "\cong"'] & \pi_1(S^1) \arrow[u, "\mathrm{id}"'] \arrow[r] \arrow[d, "\mathrm{id}"'] & \pi_1(j^*E) \arrow[u] \arrow[d] \\ \pi_2(\mathbb{C}P^n) \arrow[r, "\partial_3"] & \pi_1(S^1) \arrow[r] & \pi_1(S^{2n+1}) \end{tikzcd}$$

The vertical map on the lower left is an isomorphism, because both $\mathbb{C}P^n$ and $\mathbb{C}P^n - D^{2n}$ are simply connected and so by the Hurewicz Theorem

$$\begin{tikzcd} \pi_2(\mathbb{C}P^n - D^{2n}) \arrow[r] \arrow[d, "\cong"] & \pi_2(\mathbb{C}P^n) \arrow[d, "\cong"] \\ H_2(\mathbb{C}P^n - D^{2n};\mathbb{Z}) \arrow[r, "\cong"] & H_2(\mathbb{C}P^n;\mathbb{Z}) \end{tikzcd}$$

commutes and the vertical maps are isomorphisms. We now claim that ∂_1 is surjective. For this it suffices to show surjectivity of ∂_2, which in turn is equivalent to surjectivity of ∂_3. But this is clear by exactness of the rows and $\pi_1(S^{2n+1}) = 0$. Consequently

$$\pi_1(S^1) \to \pi_1(E)$$

is the zero map and therefore we conclude from exactness of

$$\pi_1(S^1) \to \pi_1(E) \to \pi_1(M) \to \pi_0(S^1),$$

that $\pi_1(E) \to \pi_1(M)$ is indeed an isomorphism. $\square$

The above construction now gives examples for circle bundles with positive scalar curvature over enlargeable manifolds in all even dimensions ≥ 4. To also get examples in odd dimensions let $E \to M$ be as above and let $\pi\colon M \times S^1 \to M$ be the projection. Then the pullback bundle $\pi^*E \cong E \times S^1$ clearly admits a metric of positive scalar curvature and by Lemma 1.2 the base space $M \times S^1$ is still enlargeable.